ORCA: Ocean Resource & Cognitive Agent

System Architecture, Technology Stack & How It Works (Deep Technical Guide)

Problem Statement ID: SIH26176	**System: ORCA Marine Platform**
Team: CodeCatalysts	**Target: SIH Evaluation & Technical Assessment**
Core Backend: FastAPI / Python 3.11 / PostGIS	**Core Frontend: React 19 / Vite / HTML5 Canvas PWA**

1. High-Level System Architecture

ORCA is engineered across four modular, decoupled operational tiers designed to guarantee zero-latency local fallback and seamless offline capabilities at deep sea:

Architectural Tier	Technologies	Functional Scope & Capabilities
Client Cockpit Layer	React 19 SPA + Vanilla PWA	Interactive marine map stage with 60 FPS HTML5 Canvas fluid vector simulation (winds, waves, currents). Operates offline via Service Worker & local cache.
API Gateway & Security	FastAPI / ASGI / Uvicorn	High-throughput async endpoint layer (/api/chat, /api/safety, /api/pfz, /api/route, /api/emergency). Enforces Pydantic v2 validation and rate-limiting.
Agentic Orchestration	LangGraph Multi-Agent Engine	Coordinated graph of 8 specialized agents: Planning, Data Retrieval, Ocean Analytics, Weather, Alert, Risk Assessment, Geospatial, and Vernacular Synthesis.
Scientific Math Engines	Deterministic Python Math	Zero-hallucination computational core: Risk Matrix (0-100), A* Nautical Geodesic Routing with reef/MPA avoidance, and INCOIS PFZ Thermal Front Edge Detection.
Live Ingestion Connectors	Async Polling & Webhooks	Automated parallel data pipelines to INCOIS OMNI buoys, NOAA GHRSST L4 satellite rasters, IMD Damini lightning warnings, and AIS marine traffic.
Distress & GMDSS Subsystem	Coast Guard MRCC Integration	Direct SOS beacon routing to MRCC Kochi, Mumbai, Chennai, and Port Blair with cryptographically signed SHA-256 tokens and VHF Ch 16 / DSC Ch 70 protocols.

2. Comprehensive Technology Stack

Domain / Tier	Selected Technology	Role & Rationale
Backend Framework	Python 3.11 / FastAPI / Uvicorn	Microsecond async routing, auto OpenAPI documentation, non-blocking execution.
Agent Orchestration	LangGraph / Custom StateGraph	Deterministic agent state management, graph cycles, and multi-turn conversational context.
AI Intent & Planning	Claude 3.5 Sonnet / Heuristics	Intent classification and task decomposition with zero-hallucination local fallback.
Spatial Geometry	Ray-Casting Algorithm & PostGIS	Sub-millisecond point-in-polygon containment for EEZ, IMBL, and Marine Protected Areas.
Data Validation	Pydantic v2	Strict runtime typing and schema serialization for maritime telemetry and buoy data.
Web Application	React 19 / Vite 8 / Custom CSS	High-performance reactive frontend with interactive resizable panels and responsive layout.
PWA & Offline Core	Vanilla ES6+ / Service Worker	Loads in under 400ms on low-spec marine tablets; caches vector tiles and coastal ports.
Fluid Ocean Canvas	HTML5 2D Canvas Engine	Runs locked 60 FPS particle animations for wind vectors, swell waves, and currents.
Mapping & GIS	Leaflet.js 1.9.4 / Mapbox tiles	Interactive nautical charts with draggable waypoints, bearing indicators, and hazard overlays.
Voice & Vernacular	Web Speech API (STT & TTS)	Zero-click voice queries and spoken advisories across 20 Indian coastal languages.

Automated Testing	Pytest 9.1 (34 Automated Tests)	100% test pass rate across unit tests, API integration, and SIH scenario benchmarks.

3. How It Works: End-to-End Operational Workflow

Stage 1: Intent & Location Ingestion: The user speaks or types a query (e.g., "Is it safe to fish tuna 15 NM off Cochin?"). The system extracts craft type, home port, and target parameters. If no port is spoken, it automatically reconciles coordinates from active GPS or map clicks, referencing an embedded database of 35+ coastal ports.

Stage 2: Parallel Multi-Agent Data Retrieval: The Marine Data Retrieval Agent fires parallel asynchronous requests to INCOIS live buoys, NOAA GHRSST L4 satellite rasters, Open-Meteo wind/wave models, and IMD cyclone warnings. Every piece of data is stamped with an auditable provenance ledger detailing satellite sensor, retrieval time, and freshness latency.

Stage 3: Deterministic Risk Assessment (Zero-Hallucination): The Risk Assessment Engine evaluates conditions using an objective mathematical formula: Risk Score $R = \min(100, R_{wave} + R_{wind} + R_{hazard} + R_{geofence} + R_{freshness})$. Significant wave height >= 3.5m adds +45 pts; gale winds >= 34 kts add +40 pts; active cyclones add +45 pts; stale data (>12h) adds an explicit +8 pt uncertainty penalty. Produces decisive verdicts: Safe to Sail (<=25), Caution (26-47), Unfavourable (48-67), or Prohibited (>=68).

Stage 4: PFZ Satellite Analytics & Thermal Front Detection: The Ocean Analytics & PFZ Engine analyzes thermal gradients (Delta T / km >= 0.5 deg C/km) and chlorophyll-a concentrations (>= 1.0 mg/m3) to locate active upwelling fronts where pelagic species (Yellowfin Tuna, Mackerel, Sardine) congregate. Automatically calculates dynamic bearing, nautical distance, and fuel burn.

Stage 5: Autonomous Nautical Routing & Land Bypass: The Route Engine calculates safe nautical passages using Great-Circle and A* pathfinding. Between the West Coast (Arabian Sea) and East Coast (Bay of Bengal), the Cape Comorin TSS Rounding protocol forces waypoints through 7.65 deg N, 77.40 deg E and Wadge Bank, guaranteeing 0% land intersection across the Indian peninsula while bypassing Marine Protected Areas.

Stage 6: Multilingual Synthesis & Map Synchronization: The Response Synthesis Agent translates recommendations into the user's dialect across 20 coastal languages (Tamil, Malayalam, Marathi, Bengali, Telugu, Hindi, etc.) and synchronizes vector hotspots, steering vectors, and wave contours onto the interactive map HUD.

Stage 7: Offline Operation & Marine Radio Fallback: When vessels travel past cellular range (10-15 NM offshore), the system transitions to offline PWA mode using cached SQLite vector tiles. In emergencies, the GMDSS module generates IMO-compliant distress scripts and dispatches coordinates to the nearest Indian Coast Guard MRCC with SHA-256 tokens.